

infiltrate and then control a network of devices. Try to recollect what happened last year when different hackers breached the Uconnect system of Jeep Cherokee, and controlled everything, right from the car's air conditioning system to its accelerator. If we have a perfect match between IoT and open source, such threats can be limited to a great extent.

- In order to ensure adequate levels of security before we embark on our IoT journey, we must define our security guidelines to cover cross-authentication-methods between devices and servers communicating with these devices as well as proactive malignant code detection. However, such security risks are unfortunately made worse by the often opaque security vulnerabilities within the open source software. It is essential that we give our developers easily accessible open source usage and vulnerability information continuously throughout our software development life cycles so that they can continuously keep a vulnerability check.
- Almost every device that is capable of connecting to a network runs at least a primitive operating system (along with the code) which makes it function. If we have an open source code base for this, then it allows the security of the device to be inspected, tested and when required, easily patched to help keep the intruders out. Different secure operating systems, such as the Linux kernel, can be optimised for the embedded devices to help keep data as well as devices safe.

Applications of IoT

1. IoT helps in energy management by integrating the sensing and actuation systems connected to the Internet.
2. The IoT intelligent systems help in the rapid manufacture of new products, dynamically responding to different product demands. They enable the real-time optimisation of the manufacturing and supply chain networks, by networking sensors, machinery and the control systems.
3. IoT typically makes use of sensors to assist in environmental protection by monitoring air or water quality, soil or atmospheric conditions, and can even include different areas like monitoring the movement of wildlife as well as their habitat.
4. IoT devices can also be used to monitor and control the electrical, mechanical and electronic systems used in various types of buildings.
5. IoT is capable of monitoring and controlling infrastructure like railway tracks, bridges, on- and offshore wind farms, etc.
6. IoT can also be used to enable emergency notification systems and remote health monitoring. Health monitoring devices can range from heart rate and blood pressure monitors to other advanced devices capable of monitoring specialised implants.
7. IoT can assist us in integrating communications, controls and information processing across various transportation systems as well.

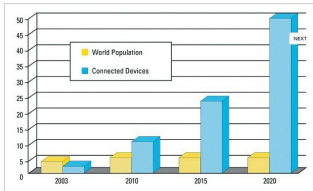


Figure 5: Growth in human population and IoT connected devices (Data source: Cisco)



Figure 6: Quick facts about IoT (Image source: googleimages.com)

8. A large number of IoT devices such as wearables, connected cars, residences, entertainment and smart homes are being developed for consumer use as well.
9. The IoT system is also very useful in reaching out to targeted sections of a population in order to convey a message that's important for them instead of using various older methods such as newspapers. END

References

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